Journal of the Physical Society of Japan
Vol. 68, No. 5, May, 1999, pp. 1508-1512

Darboux Transformation and Soliton Solutions for Generalized Nonlinear Schrödinger Equations

Xianguo GENG and Hon-Wah TAM[1]

Department of Mathematics, Zhengzhou University, Zhengzhou, Henan 450052, People's Republic of China
[1]*Department of Computer Science, Hong Kong Baptist University, 224 Waterloo Road, Kowloon, Hong Kong, People's Republic of China*

(Received November 19, 1998)

A Darboux transformation with multi-parameters for coupled generalized nonlinear Schrödinger equations is derived with the help of a gauge transformation of the spectral problem. As a reduction, a Darboux transformation of the generalized nonlinear Schrödinger equation is obtained, from which soliton solutions of the generalized nonlinear Schrödinger equation are given.

KEYWORDS: Darboux transformation, solitons, generalized nonlinear Schrödinger equations

§1. Introduction

The nonlinear Schrödinger equation

$$\mathrm{i}u_t + u_{xx} + 2|u|^2u = 0, \tag{1.1}$$

arises from a wide variety of fields, such as nonlinear optics, the theory of deep water waves, and plasma physics.[1-4] In the mathematical theory of integrable equations, the discovery of the integrability of this equation by the inverse scattering transformation method has promoted to a large extent the understanding of the generality of this method and subsequently its fast development. A number of research areas on equation (1.1) has been conducted. For example, its soliton solutions, conserved quantities, Bäcklund transformation, Darboux transformation (DT), and others have been discussed in refs. 1–6.

In the ref. 7, a generalized nonlinear Schrödinger (GNLS) equation

$$\mathrm{i}u_t + u_{xx} - 2\varepsilon|u|^2u + 4\beta^2|u|^4u + 4\mathrm{i}\varepsilon\beta(|u|^2)_xu = 0, \tag{1.2}$$

has been studied by means of the gauge transformation, where β is a real constant, $\varepsilon = \pm 1$. Recently ones in a series of papers[8-10] have discussed the gauge connections between (1.2) and other soliton equations, and showed that (1.2) possesses a Lax representation, the Hamiltonian structure, and other properties.

In this letter, we are going to derive a DT[11-13] with multi-parameters for a generalized AKNS spectral problem and the associated coupled GNLS equations (see (2.1) below), from which the solutions of the coupled GNLS equations are reduced to solving a linear algebraic system and two first-order ordinary differential equations. As a reduction, a DT of the GNLS equation (1.2) is obtained. A systematic procedure is given in detail to solve the GNLS equation. The present paper is organized as follows. In §2, we shall construct a DT of the generalized AKNS spectral problem resorting to the gauge transformation between spectral problems. The DT is further proved to be also the DT of the coupled GNLS equations under certain conditions. In §3, a DT of the GNLS equation (1.2) is discussed through the reduction technique. Furthermore, one- and two-soliton solutions of the GNLS equation (1.2) are explicitly given by the application of its DT.

§2. Darboux Transformation

In this section, we shall construct a DT of the coupled GNLS equations

$$\begin{aligned} &\mathrm{i}u_t + u_{xx} - 2u^2v + 4\beta^2u^3v^2 + 4\mathrm{i}\beta(uv)_xu = 0,\\ &\mathrm{i}v_t - v_{xx} + 2uv^2 - 4\beta^2u^2v^3 + 4\mathrm{i}\beta(uv)_xv = 0. \end{aligned} \tag{2.1}$$

Equations (2.1) are reduced to the GNLS equation (1.2) when $v = \varepsilon u^*$. These equations have a Lax pair, the generalized AKNS spectral problem

$$\begin{aligned} &y_x = Uy, \quad y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix},\\ &U = \begin{pmatrix} -\mathrm{i}\zeta - \mathrm{i}\beta uv & u \\ v & \mathrm{i}\zeta + \mathrm{i}\beta uv \end{pmatrix}, \end{aligned} \tag{2.2}$$

and the auxiliary problem

$$y_t = Vy, \quad V = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}, \tag{2.3}$$

with

$$\begin{aligned} &a = -2\mathrm{i}\zeta^2 + a_1, \quad b = 2u\zeta + b_1, \quad c = 2v\zeta + c_1,\\ &a_1 = \beta(u_xv - uv_x) + 4\mathrm{i}\beta^2u^2v^2 - \mathrm{i}uv,\\ &b_1 = \mathrm{i}u_x - 2\beta u^2v, \quad c_1 = -\mathrm{i}v_x - 2\beta uv^2, \end{aligned}$$

where u and v are two potentials, and ζ is a constant spectral parameter.

In what follows, we shall construct the DT of the spectral problem (2.2). Let $\phi = (\phi_1, \phi_2)^T, \psi = (\psi_1, \psi_2)^T$ be two basic solutions of (2.2) and use (ϕ, ψ) to define a

linear algebraic system

$$\sum_{k=0}^{N-1}(A_k+B_k\alpha_j)\zeta_j^k=-\zeta_j^N,\quad 1\le j\le 2N, \tag{2.4a}$$

$$\sum_{k=0}^{N-1}(C_k+D_k\alpha_j)\zeta_j^k=-\alpha_j\zeta_j^N,\quad 1\le j\le 2N, \tag{2.4b}$$

with

$$\alpha_j=\frac{\phi_2(\zeta_j)-\gamma_j\psi_2(\zeta_j)}{\phi_1(\zeta_j)-\gamma_j\psi_1(\zeta_j)},\quad 1\le j\le 2N, \tag{2.5}$$

where parameters ζ_j and γ_j ($\zeta_k\neq\zeta_j,\gamma_k\neq\gamma_j,k\neq j$) are suitably chosen such that the determinants of the coefficients for (2.4a) and (2.4b) are nonzero. Hence, A_k, B_k, C_k, and D_k ($0\le k\le N-1$) are uniquely determined by (2.4). Now we introduce a gauge transformation of the spectral problem (2.2)

$$\hat{y}=Ty,\quad T=\begin{pmatrix}A & B\\ C & D\end{pmatrix}, \tag{2.6}$$

with

$$A=A_N\left(\zeta^N+\sum_{j=0}^{N-1}A_j\zeta^j\right),\quad B=A_N\sum_{j=0}^{N-1}B_j\zeta^j,$$

$$C=\frac{1}{A_N}\sum_{j=0}^{N-1}C_j\zeta^j,\quad D=\frac{1}{A_N}\left(\zeta^N+\sum_{j=0}^{N-1}D_j\zeta^j\right),$$

where A_N will be determined later. From (2.4) we have

$$\left(\zeta_j^N+\sum_{k=0}^{N-1}A_k\zeta_j^k\right)\left(\zeta_j^N+\sum_{k=0}^{N-1}D_k\zeta_j^k\right)=\sum_{k=0}^{N-1}\sum_{l=0}^{N-1}B_kC_l\zeta_j^{k+l},$$

that is

$$\zeta_j^{2N}+\sum_{k=0}^{N-1}(A_k+D_k)\zeta_j^{N+k}+\sum_{k=0}^{N-1}\sum_{l=0}^{N-1}(A_kD_l-B_kC_l)\zeta_j^{k+l}=0. \tag{2.7}$$

Here the gauge transformation means that it transforms a spectral problem into another spectral problem, that is, it is a transformation between two distinct spectral problems. Equation (2.7) implies that ζ_j ($1\le j\le 2N$) are $2N$ roots of $(2N)$th-order polynomial $\det T$. Therefore, we have

$$\det T=\prod_{j=1}^{2N}(\zeta-\zeta_j). \tag{2.8}$$

Under the gauge transformation (2.6), the spectral problem (2.2) is transformed into the following spectral problem of $\hat{y}$ when $\zeta\neq\zeta_j$ ($1\le j\le 2N$):

$$\hat{y}_x=\hat{U}\hat{y}, \tag{2.9}$$

where

$$\hat{U}=(T_x+TU)T^{-1}. \tag{2.10}$$

It turns out that $\zeta=\zeta_j$ ($1\le j\le 2N$) are removable isolated singularities of $\hat{U}$ (see (2.12) below). Thus we can define $\hat{U}$ for all ζ by analytic continuation.

A gauge transformation of a spectral problem is called a Darboux transformation of the spectral problem if it transforms the spectral problem into another spectral problem of the same type.

Theorem 1. Let A_N satisfy a first-order ordinary differential equation

$$\partial_x\ln A_N=2\beta(vB_{N-1}-uC_{N-1}-2\mathrm{i}B_{N-1}C_{N-1}). \tag{2.11}$$

Then the matrix $\hat{U}$ determined by (2.10) has the same form as U:

$$\hat{U}=\begin{pmatrix}-\mathrm{i}\zeta-\mathrm{i}\beta\hat{u}\hat{v} & \hat{u}\\ \hat{v} & \mathrm{i}\zeta+\mathrm{i}\beta\hat{u}\hat{v}\end{pmatrix}, \tag{2.12}$$

where the transformation formulae from the old potentials u and v into new ones are given by

$$\hat{u}=uA_N^2+2\mathrm{i}A_N^2B_{N-1},\quad \hat{v}=\frac{v}{A_N^2}-2\mathrm{i}\frac{C_{N-1}}{A_N^2}. \tag{2.13}$$

The transformation (2.6) and (2.13): $(y;u,v)\to(\hat{y};\hat{u},\hat{v})$ is a DT of the spectral problem (2.2).

Proof. Let $T^{-1}=T^*/\det T$ and

$$(T_x+TU)T^*=\begin{pmatrix}f_{11}(\zeta) & f_{12}(\zeta)\\ f_{21}(\zeta) & f_{22}(\zeta)\end{pmatrix}. \tag{2.14}$$

It is easy to see that $f_{11}(\zeta),f_{22}(\zeta)$ or $f_{12}(\zeta),f_{21}(\zeta)$ are $(2N+1)$th-order polynomials in ζ or $(2N)$th-order polynomials in ζ, respectively. By using (2.5), (2.2), and (2.4), we obtain

$$\alpha_{jx}=v-u\alpha_j^2+2(\mathrm{i}\zeta_j+\mathrm{i}\beta uv)\alpha_j,\quad 1\le j\le 2N, \tag{2.15}$$

$$A(\zeta_j)=-\alpha_jB(\zeta_j),\quad C(\zeta_j)=-\alpha_jD(\zeta_j). \tag{2.16}$$

Resorting to (2.15) and (2.16), it can be verified that ζ_j ($1\le j\le 2N$) are roots of $f_{kj}(\zeta)$, $k,j=1,2$. Therefore, we have from (2.8)

$$(T_x+TU)T^*=(\det T)P(\zeta), \tag{2.17}$$

with

$$P(\zeta)=\begin{pmatrix}p_{11}^{(1)}\zeta+p_{11}^{(0)} & p_{12}^{(0)}\\ p_{21}^{(0)} & p_{22}^{(1)}\zeta+p_{22}^{(0)}\end{pmatrix},$$

where $p_{kj}^{(l)}$ are independent of ζ. Equation (2.17) can be written as

$$T_x+TU=P(\zeta)T. \tag{2.18}$$

Equating the coefficients of ζ^{N+1} and ζ^N in (2.18), we find

$$p_{11}^{(1)}=-i,\quad p_{22}^{(1)}=i,\quad p_{12}^{(0)}=\hat{u},\quad p_{21}^{(0)}=\hat{v}, \tag{2.19}$$

$$p_{11}^{(0)}=-p_{22}^{(0)}=-\mathrm{i}\beta uv+\partial_x\ln A_N. \tag{2.20}$$

Substituting (2.11) into (2.20) yields

$$p_{11}^{(0)}=-p_{22}^{(0)}=-\mathrm{i}\beta(u+2\mathrm{i}B_{N-1})(v-2\mathrm{i}C_{N-1})=-\mathrm{i}\beta\hat{u}\hat{v}. \tag{2.21}$$

The proof is thus completed.

Let the two solutions ϕ and ψ of (2.2) satisfy equation (2.3) as well. Differentiating $\hat{y} = Ty$ with respect to t yields

$$\hat{y}_t = \hat{V}\hat{y}, \quad \hat{V} = (T_t + TV)T^{-1}. \tag{2.22}$$

Then we can prove the following assertion.

Theorem 2. Suppose that the time dependence of A_N obeys the first-order ordinary differential equation with respect to the variable t

$$\begin{aligned}&\partial_t \ln A_N\\ &= 2\mathrm{i}\beta(uC_{N-1,x} - u_xC_{N-1} + vB_{N-1,x} - v_xB_{N-1})\\ &\quad - 16\beta^2uv(vB_{N-1} - uC_{N-1} - 2\mathrm{i}B_{N-1}C_{N-1})\\ &\quad + 4\beta(C_{N-1}B_{N-1,x} - B_{N-1}C_{N-1,x}).\end{aligned} \tag{2.23}$$

Then the matrix $\hat{V}$ defined by the second expression of (2.22) has the same form as V, in which the old potentials u and v are mapped into $\hat{u}$ and $\hat{v}$ according to the same DT (2.13).

Proof. Let $T^{-1} = T^*/\det T$ and

$$(T_t + TV)T^* = \begin{pmatrix} g_{11}(\zeta) & g_{12}(\zeta) \\ g_{21}(\zeta) & g_{22}(\zeta) \end{pmatrix}. \tag{2.24}$$

A direct calculation shows that $g_{11}(\zeta)$ and $g_{22}(\zeta)$ are $(2N+2)$th-order polynomials in ζ, $g_{12}(\zeta)$ and $g_{21}(\zeta)$ are $(2N+1)$th-order polynomials in ζ. By virtue of (2.5), (2.3), and (2.16), we have

$$\begin{aligned}\alpha_{jt} &= c(\zeta_j) - 2a(\zeta_j)\alpha_j - b(\zeta_j)\alpha_j^2,\\ A_t(\zeta_j) &= -B_t(\zeta_j)\alpha_j - B(\zeta_j)\alpha_{jt},\\ C_t(\zeta_j) &= -D_t(\zeta_j)\alpha_j - D(\zeta_j)\alpha_{jt}.\end{aligned} \tag{2.25}$$

It can be easily verified by (2.25) and (2.16) that ζ_j $(1 \le j \le 2N)$ are roots of $g_{kj}(\zeta)$, $k, j = 1, 2$. Therefore, we arrive at

$$(T_t + TV)T^* = (\det T)Q(\zeta),$$

with

$$Q(\zeta) = \begin{pmatrix} q_{11}^{(2)}\zeta^2 + q_{11}^{(1)}\zeta + q_{11}^{(0)} & q_{12}^{(1)}\zeta + q_{12}^{(0)} \\ q_{21}^{(1)}\zeta + q_{21}^{(0)} & q_{22}^{(2)}\zeta^2 + q_{22}^{(1)}\zeta + q_{22}^{(0)} \end{pmatrix}$$

that is

$$T_t + TV = Q(\zeta)T. \tag{2.26}$$

By comparing the coefficients of ζ^{N+2}, ζ^{N+1}, and ζ^N in (2.26), we get

$$\begin{aligned}&q_{11}^{(2)} = -q_{22}^{(2)} = -2\mathrm{i}, \quad q_{11}^{(1)} = q_{22}^{(1)} = 0,\\ &q_{12}^{(1)} = 2\hat{u}, \quad q_{21}^{(1)} = 2\hat{v},\\ &q_{12}^{(0)} = (b_1 + 4\mathrm{i}B_{N-2} + 2uA_{N-1} - 2\hat{u}A_N^{-2}D_{N-1})A_N^2,\\ &q_{21}^{(0)} = (c_1 - 4\mathrm{i}C_{N-2} + 2vD_{N-1} - 2\hat{v}A_N^2A_{N-1})A_N^{-2},\end{aligned} \tag{2.27}$$

$$\begin{aligned}&A_{Nt} - q_{11}^{(0)}A_N = -(a_1 + 2vB_{N-1})A_N + 2\hat{u}A_N^{-1}C_{N-1},\\ &A_{Nt} + q_{22}^{(0)}A_N = -(a_1 - 2uC_{N-1})A_N - 2\hat{v}A_N^3B_{N-1}.\end{aligned} \tag{2.28}$$

$$\begin{aligned}&2\mathrm{i}B_{N-2} + uA_{N-1} - \hat{u}A_N^{-2}D_{N-1}\\ &= -B_{N-1,x} - B_{N-1}[\partial_x \ln A_N + \mathrm{i}\beta(uv + \hat{u}\hat{v})],\\ &\quad -2\mathrm{i}C_{N-2} + vD_{N-1} - \hat{v}A_N^2A_{N-1}\\ &= -C_{N-1,x} + C_{N-1}[\partial_x \ln A_N + \mathrm{i}\beta(uv + \hat{u}\hat{v})].\end{aligned} \tag{2.29}$$

Equating the coefficients of ζ^{N-1} in (2.18) and noticing (2.19)–(2.21) yields

By using (2.21) and (2.20), we have

$$\beta\hat{u}\hat{v} = \beta uv + \mathrm{i}\partial_x \ln A_N. \tag{2.30}$$

Substituting (2.30) into (2.28) and noticing (2.31), we get

$$\begin{aligned}q_{12}^{(0)} &= (b_1 - 2B_{N-1,x} - 4\mathrm{i}\beta uvB_{N-1})A_N^2,\\ q_{21}^{(0)} &= (c_1 - 2C_{N-1,x} + 4\mathrm{i}\beta uvC_{N-1})A_N^{-2}.\end{aligned} \tag{2.31}$$

A direct calculation shows by (2.13) and (2.31) that

$$\begin{aligned}\mathrm{i}\hat{u}_x - 2\beta\hat{u}^2\hat{v} &= (b_1 - 2B_{N-1,x} - 4\mathrm{i}\beta uvB_{N-1})A_N^2,\\ -\mathrm{i}\hat{v}_x - 2\beta\hat{u}\hat{v}^2 &= (c_1 - 2C_{N-1,x} + 4\mathrm{i}\beta uvC_{N-1})A_N^{-2},\end{aligned} \tag{2.32}$$

which, together with (2.32), implies that

$$q_{12}^{(0)} = \mathrm{i}\hat{u}_x - 2\beta\hat{u}^2\hat{v}, \quad q_{21}^{(0)} = -\mathrm{i}\hat{v}_x - 2\beta\hat{u}\hat{v}^2. \tag{2.33}$$

From (2.13) and (2.29), we have

$$\begin{aligned}&\partial_t \ln A_N - q_{11}^{(0)}\\ &= -a_1 - 2vB_{N-1} + 2uC_{N-1} + 4\mathrm{i}B_{N-1}C_{N-1},\\ &\qquad q_{22}^{(0)} = -q_{11}^{(0)}.\end{aligned} \tag{2.34}$$

Through tedious calculations, we obtain from (2.13), (2.11), and (2.31) that

$$\begin{aligned}&\beta(\hat{u}_x\hat{v} - \hat{u}\hat{v}_x) + 4\mathrm{i}\beta^2\hat{u}^2\hat{v}^2 - \mathrm{i}\hat{u}\hat{v}\\ &\quad = a_1 + 2\mathrm{i}\beta(uC_{N-1,x} - u_xC_{N-1} + vB_{N-1,x} - v_xB_{N-1})\\ &\quad\quad + 2(1 - 8\beta^2uv)(vB_{N-1} - uC_{N-1} - 2\mathrm{i}B_{N-1}C_{N-1})\\ &\quad\quad + 4\beta(C_{N-1}B_{N-1,x} - B_{N-1}C_{N-1,x}).\end{aligned} \tag{2.35}$$

(2.36)

(2.37)

Substituting (2.23) into (2.35) and using (2.36) yields

$$q_{11}^{(0)} = -q_{22}^{(0)} = \beta(\hat{u}_x\hat{v} - \hat{u}\hat{v}_x) + 4\mathrm{i}\beta^2\hat{u}^2\hat{v}^2 - \mathrm{i}\hat{u}\hat{v}. \tag{2.38}$$

This completes the proof.

According to theorems 1 and 2, the transformation (2.6) and (2.13) transforms the Lax pair (2.2) and (2.3) into another Lax pair of the same type (2.9) and (2.22).

Therefore, both of the Lax pairs lead to the same coupled GNLS equations (2.1). The transformation (2.13) is also called a DT of the coupled GNLS equations (2.1). We get immediately the following fact.

Theorem 3. Every solution (u, v) of (2.1) is mapped into a new solution $(\hat{u}, \hat{v})$ of (2.1) under the DT (2.13), where B_{N-1} and C_{N-1} are given by the linear algebraic system (2.4), and A_N is determined by the two first-order ordinary differential equations (2.11) and (2.23).

Remark If $\beta = 0$, eqs. (2.1) are reduced to the coupled nonlinear Schrödinger equations. From (2.11) and (2.23), we have $A_N = 1$ in this particular case. All the above results still hold.

§3. The Reduction of Darboux Transformation and Application

In this section, we shall discuss the DT of the GNLS equation (1.2) and give its explicit solutions. To this end, we always assume that $v = \varepsilon u^*$. In this case we choose two solutions of (2.2) and (2.3):

$$\begin{aligned} \phi(\zeta) &= (\phi_1(\zeta), \phi_2(\zeta))^T, \\ \psi(\zeta) &= (\varepsilon\phi_2^*(\zeta^*), \phi_1^*(\zeta^*))^T, \end{aligned} \tag{3.1}$$

and parameters

$$\zeta_{2j} = \zeta_{2j-1}^*, \quad \gamma_{2j} = \varepsilon\gamma^{*-1}_{2j-1}, \quad 1 \le j \le N. \tag{3.2}$$

It is easy to prove that $\alpha_{2j}^{-1} = \varepsilon\alpha_{2j-1}^*$, $D_k = A_k^*$, $C_k = \varepsilon B_k^*$, $(1 \le j \le N, 0 \le k \le N-1)$. Here α_{2j-1} is given by

$$\alpha_{2j-1} = \frac{\phi_2(\zeta_{2j-1}) - \gamma_{2j-1}\phi_1^*(\zeta_{2j-1}^*)}{\phi_1(\zeta_{2j-1}) - \varepsilon\gamma_{2j-1}\phi_2^*(\zeta_{2j-1}^*)}, \tag{3.3}$$

and A_k and B_k are determined by the following linear algebraic system

$$\begin{aligned} &\sum_{k=0}^{N-1}(A_k + \alpha_{2j-1}B_k)\zeta_{2j-1}^k = -\zeta_{2j-1}^N, \\ &\sum_{k=0}^{N-1}(\alpha_{2j-1}^*A_k + \varepsilon B_k)\zeta^{*k}_{2j-1} \\ &\qquad = -\alpha_{2j-1}^*\zeta^{*N}_{2j-1}, \quad 1 \le j \le N. \end{aligned} \tag{3.4}$$

Therefore, we prove easily the following assertion.

Theorem 4. Let A_N be a solution of the first-order differential equations

$$\begin{aligned} \partial_x \ln A_N &= 4\mathrm{i}\varepsilon\beta\left[\mathrm{Im}\,(u^*B_{N-1}) - |B_{N-1}|^2\right], \\ \partial_t \ln A_N &= 4\mathrm{i}\varepsilon\beta\,\mathrm{Re}\,(u^*B_{N-1,x} - u_x^*B_{N-1}) \\ &\quad +8\mathrm{i}\varepsilon\beta\,\mathrm{Im}\,(B_{N-1}^*B_{N-1,x}) \\ &\quad -32\mathrm{i}\beta^2|u|^2\left[\mathrm{Im}\,(u^*B_{N-1}) - |B_{N-1}|^2\right], \end{aligned} \tag{3.5}$$

and $|A_N| = 1$, where Re and Im stand for the real and imaginary parts. respectively. Then the Darboux transformation

$$\hat{u} = uA_N^2 + 2\mathrm{i}A_N^2B_{N-1}, \tag{3.6}$$

is a solution of the GNLS equation (1.2). Here the condition $|A_N| = 1$ means that constant of integration for A_N is selected to be purely imaginary.

In the following, we shall apply the DT to give explicit solutions of the GNLS equation

$$\mathrm{i}u_t + u_{xx} + 2|u|^2u + 4\beta^2|u|^4u - 4\mathrm{i}\beta(|u|^2)_xu = 0, \tag{3.7}$$

which is a special case of (1.2) with $\varepsilon = -1$. Substituting the trivial solution $u = 0$, $(v = -u = 0)$, of (3.7) into (2.2) and (2.3), two basic solutions $\phi(\zeta)$ and $\psi(\zeta)$ are chosen as

$$\phi(\zeta) = \begin{pmatrix} \exp(-\mathrm{i}\zeta x - 2\mathrm{i}\zeta^2 t) \\ 0 \end{pmatrix},$$

$$\psi(\zeta) = \begin{pmatrix} 0 \\ \exp(\mathrm{i}\zeta x + 2\mathrm{i}\zeta^2 t) \end{pmatrix},$$

according to (3.1). Equations (3.3)–(3.5) can be written as

$$\alpha_{2j-1} = -\exp(2\mathrm{i}\zeta_{2j-1}x + 4\mathrm{i}\zeta_{2j-1}^2t + \delta_j + \mathrm{i}\nu_j), \tag{3.8}$$

$$\begin{aligned} &\sum_{k=0}^{N-1}(A_k + \alpha_{2j-1}B_k)\zeta_{2j-1}^k = -\zeta_{2j-1}^N, \\ &\sum_{k=0}^{N-1}(\alpha_{2j-1}^*A_k - B_k)\zeta^{*k}_{2j-1} \\ &\qquad = -\alpha_{2j-1}^*\zeta^{*N}_{2j-1}, \quad 1 \le j \le N, \end{aligned} \tag{3.9}$$

$$\begin{aligned} \partial_x \ln A_N &= 4\mathrm{i}\beta|B_{N-1}|^2, \\ \partial_t \ln A_N &= -8\mathrm{i}\beta\,\mathrm{Im}\,(B_{N-1}^*B_{N-1,x}), \end{aligned} \tag{3.10}$$

where $\gamma_{2j-1} = \exp(\delta_j + \mathrm{i}\nu_j)$. For simplicity, we shall discuss the two special cases $N = 1$ and $N = 2$.

(I) For $N = 1$, let $\zeta_1 = \xi_1 + \mathrm{i}\eta_1$ $(\xi_1 > 0, \eta_1 > 0, \xi_1 \ne \eta_1)$. Then we obtain from the linear algebraic system (3.9) that

$$\begin{aligned} B_0 = \mathrm{i}\eta_1 &\exp\{-\mathrm{i}[2\xi_1x + 4(\xi_1^2 - \eta_1^2)t + \nu_1]\} \\ &\times\mathrm{sech}\,(2\eta_1x + 8\xi_1\eta_1t - \delta_1), \end{aligned} \tag{3.11}$$

which together with system (3.10) implies that

$$\begin{aligned} A_1 = \exp\{\mathrm{i}[\rho_1 &+ 2\beta\eta_1\exp(2\eta_1x + 8\xi_1\eta_1t - \delta_1) \\ &\times\mathrm{sech}(2\eta_1x + 8\xi_1\eta_1t - \delta_1)]\}, \end{aligned} \tag{3.12}$$

where ρ_1 is a real constant of integration. A one-soliton solution of the GNLS equation (3.7) is obtained with the help of the DT (3.6):

$$\hat{u} = -2\eta_1\exp[\mathrm{i}f(x,t)]\,\mathrm{sech}\,(2\eta_1x + 8\xi_1\eta_1t - \delta_1), \tag{3.13}$$

where

$$\begin{aligned} f(x,t) = &-2\xi_1x - 4(\xi_1^2 - \eta_1^2)t - \nu_1 + 2\rho_1 + 4\beta\eta_1 \\ &\times\exp(2\eta_1x + 8\xi_1\eta_1t - \delta_1) \\ &\times\mathrm{sech}\,(2\eta_1x + 8\xi_1\eta_1t - \delta_1). \end{aligned}$$

Plots of this solution are given in Fig. 1.

(II) For $N = 2$, let $\zeta_1 = \xi_1 + \mathrm{i}\eta_1$, $\zeta_3 = -\xi_1 + \mathrm{i}\eta_1$ $(\xi_1 > 0, \eta_1 > 0, \xi_1 \ne \eta_1)$. Then we have

$$\alpha_1 = -\exp(X_1 + \mathrm{i}Y_1), \quad \alpha_3 = -\exp(X_2 + \mathrm{i}Y_2), \tag{3.14}$$

where

$$\begin{aligned} X_1 &= -2\eta_1x - 8\xi_1\eta_1t + \delta_1, \\ X_2 &= -2\eta_1x + 8\xi_1\eta_1t + \delta_2, \quad (\delta_1 \ne \delta_2), \end{aligned}$$

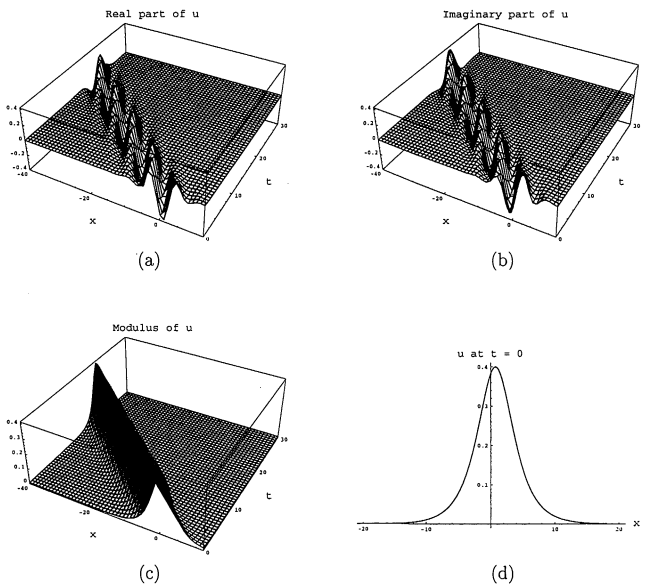


Fig. 1. One-soliton solution (3.13). $\beta = 1$, $\eta_1 = 0.2$, $\xi_1 = 0.5$, $\delta_1 = 0.3$, $\rho_1 = 0.4$, $\nu_1 = 0.6$, $\delta_2 = -0.1$, $\nu_2 = -0.2$, $\rho_2 = -0.3$.

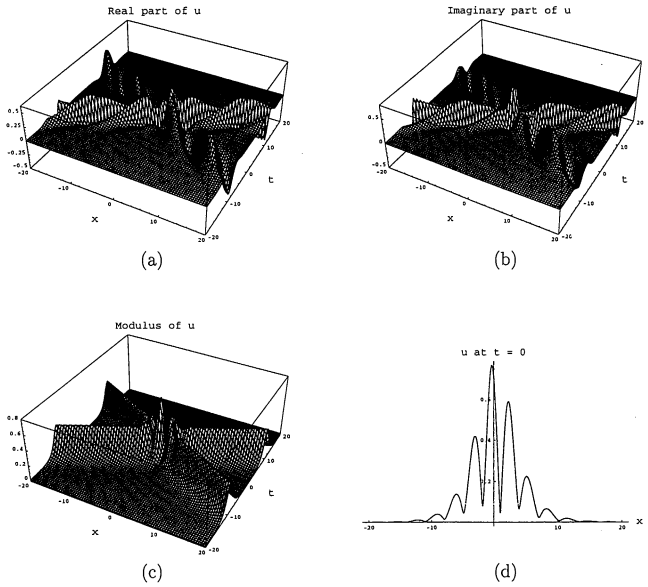


Fig. 2. Two-soliton solution (3.19). $\beta = 1$, $\eta_1 = 0.2$, $\xi_1 = 0.5$, $\delta_1 = 0.3$, $\rho_1 = 0.4$, $\nu_1 = 0.6$, $\delta_2 = -0.1$, $\nu_2 = -0.2$, $\rho_2 = -0.3$.

$$Y_1 = 2\xi_1 x + 4(\xi_1^2 - \eta_1^2)t + \nu_1,$$
$$Y_2 = -2\xi_1 x + 4(\xi_1^2 - \eta_1^2)t + \nu_2, \quad (\nu_1 \neq \nu_2).$$

Through tedious calculations, we arrive at

$$\begin{aligned}\Delta &= 4\xi_1^2\,[1 + \exp(2X_1)]\,[1 + \exp(2X_2)] \\ &\quad +4\eta_1^2[\exp(2X_1) + \exp(2X_2) \\ &\quad -2\exp(X_1 + X_2)\cos(Y_1 - Y_2)], \\ \Delta_{B_1} &= \omega_1 \exp(X_1 - \mathrm{i}Y_1) + \omega_2 \exp(X_2 - \mathrm{i}Y_2) \\ &\quad +\omega_3 \exp(X_1 + 2X_2 - \mathrm{i}Y_1) \\ &\quad +\omega_4 \exp(2X_1 + X_2 - \mathrm{i}Y_2),\end{aligned} \tag{3.15}$$

with

$$\omega_1 = 8\xi_1\eta_1(\eta_1 + \mathrm{i}\xi_1), \quad \omega_2 = -8\xi_1\eta_1(\eta_1 - \mathrm{i}\xi_1),$$
$$\omega_3 = -8\xi_1\eta_1(\eta_1 - \mathrm{i}\xi_1), \quad \omega_4 = 8\xi_1\eta_1(\eta_1 + \mathrm{i}\xi_1),$$

where Δ is the determinant of the coefficients for the linear algebraic system (3.9), and Δ_{B_1} is engendered from Δ by replacing its 4th column with $(-\zeta_1^2, -\zeta_3^2, -\alpha_1^*\zeta_1^{*2}, -\alpha_3^*\zeta_3^{*2})^T$. Therefore, from the linear algebraic system (3.9), we have

$$B_1 = \frac{\Delta_{B_1}}{\Delta}. \tag{3.16}$$

Substituting (3.16) into system (3.10) and solving, we obtain

$$A_2 = \exp\left[\mathrm{i}\left(\rho_2 + 4\beta\frac{\Omega}{\Delta}\right)\right], \tag{3.17}$$

where ρ_2 is a real constant of integration, and

$$\begin{aligned}\Omega &= 4\xi_1^2\eta_1[1 - \exp(2X_1 + 2X_2)] \\ &\quad +8\xi_1\eta_1^2\exp(X_1 + X_2)\sin(Y_1 - Y_2).\end{aligned} \tag{3.18}$$

Another solution of the GNLS equation (3.7) is obtained by using the DT (3.6)

$$\hat{u} = 2\mathrm{i}\exp\left[2\mathrm{i}\left(\rho_2 + 4\beta\frac{\Omega}{\Delta}\right)\right]\frac{\Delta_{B_1}}{\Delta}, \tag{3.19}$$

which is the two-soliton solution. (See Fig. 2.)

Acknowledgments

Project 19671074 was supported by National Natural Science Foundation of China. One of the authors (X.G. Geng) would like to thank the Henan Science Foundation Committee of China for financial support.

1) M. J. Ablowitz and H. Segur: *Solitons and the Inverse Scattering Transform* (SIAM, Philadelphia, 1981).
2) A. C. Newell: *Solitons in Mathematics and Physics* (SIAM, Philadelphia, 1985).
3) V. E. Zakharov and A. B. Shabat: Sov. Phys. JETP **34** (1972) 62.
4) L. D. Faddeev and L. A. Takhtajan: *Hamiltonian Methods in the Theory of Solitons* (Springer, Berlin, 1987).
5) C. H. Gu: *Soliton Theory and Its Application* (Springer, Berlin, 1995).
6) V. B. Matveev and M. A. Salle: *Darboux Transformation and Solitons* (Springer, Berlin, 1991).
7) A. Kundu: J. Math. Phys. **25** (1984) 3433.
8) S. Kakei, N. Sasa and J. Satsuma: J. Phys. Soc. Jpn. **64** (1995) 1519.
9) G. Z. Tu: *Liouville Integrability of Zero Curvature Equations*, Nonlinear Physics, Research Reports in Physics, ed. C. H. Gu, Y. S. Li and G. Z. Tu (Springer, Berlin, 1990) p. 2.
10) X. G. Geng: Physica A **180** (1992) 241.
11) D. Levi: Inverse Problems **4** (1988) 165.
12) C. H. Gu and Z. X. Zhou: Lett. Math. Phys. **32** (1994) 1.
13) Y. S. Li: J. Phys. A: Math. Gen. **29** (1996) 4187.